

IE6-DPL: REPORT LAB 01

SAMPLING AND QUANTIZATION

Team (Group number + letter as chosen in Moodle)	Team members	
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	3	

Please fill out the document electronically – including the yellow fields above – and upload it to Moodle once per team (no matter who uploads it, it will automatically be assigned to all members). Please note that the complete assignment can be found in another document!

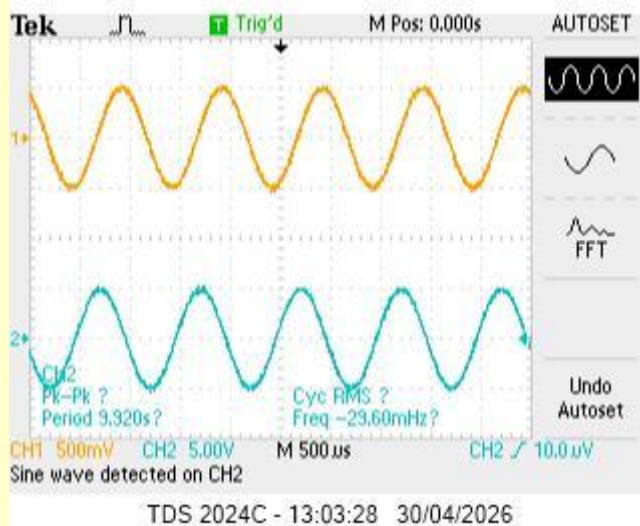
Lab task 1.1: Feeding the ADC input directly to the DAC output

In this first task, you apply a signal to the ADC and use the given program to read this signal into the DSP and output the signal at the DAC.

1. Function test of the program

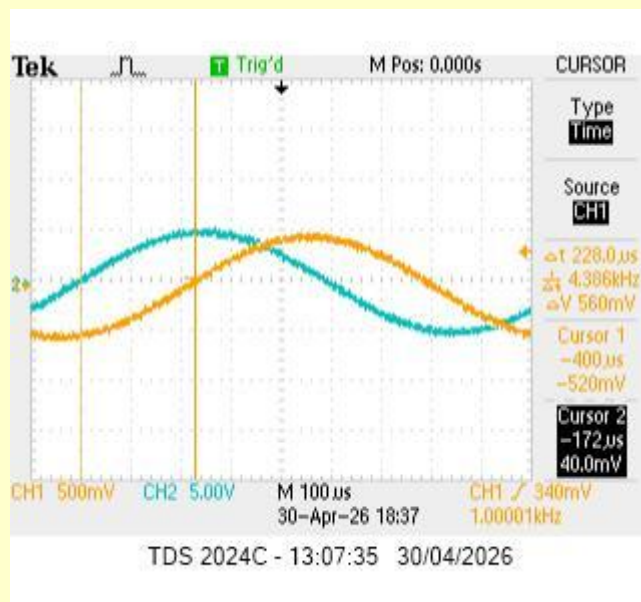
- Use the HAMEG HMF2525 function generator to apply a sinusoidal voltage to the input of the board. Mind that you have to **terminate the coax cable from the function generator with a 50 Ω resistor** as otherwise the double value of

- Determined delay value: ch1 400us
- Ch2 172us : 400us – 172us = delta t which is 228us
- Subtask 3 masked with &=0x0000, graph shows ch0 silent
- Oscilloscope screenshot: input and output signal of the



board

same



Same signal, ch1 is output, ch2 is input. Graph2: delay measurement

- Code for sub task 4 (has to be demonstrated in the lab):
- `PRU_addaRegs->dac[0] = g_i16Data[1];`
- `PRU_addaRegs->dac[0] = g_i16Data[0];`

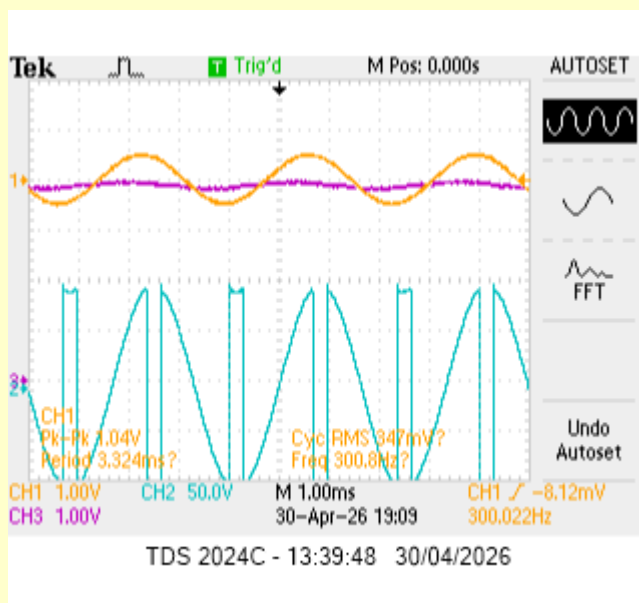
Lab task 2: Number range overflows

- Modify the DAC interrupt handler `dacInt()` that the values of both ADC inputs are multiplied by a factor scale (defined as a global variable) before they are output to the DAC outputs.
- Add the factor scale to the *Expressions window* of the CCS Debugger.
- Increase the factor scale in the *Expressions window* until you observe an overflow on the oscilloscope. **Make an oscilloscope screenshot right before and right after the overflow occurs. Specify the value of scale at which the overflow occurs and explain the signal shape in the event of an overflow in the report.**

- From which factor does number overflow (no saturation) occur?

At **21** factor the overflow occurs.

- Oscilloscope screenshot right after overflow:



Lab task 3: Quantization of speech signals

- Make sure that the audio signal is well leveled by leaving the value of factor scale as you determined it in task 2, now applied to both channels. Now increase the volume on the PC as much as possible without overflowing (you would hear this in the signal).
- Add a global variable `bitmask` to your program that manipulates both channels

```
sData[0] &= bitmask;  
sData[1] &= bitmask;
```

- Intelligibility of 1-,4-, 8-bit signal

8bit- sound clear

4- distorted understandable

1-bit unaudible

- Oscilloscope screenshot for 4-bit quantized signal